AKSHAY PARALIKAR

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EDUCATION

New York University — **09/2023 – 05/2025**
Masters of Science, Electrical Engineering — ***GPA: 4.0/4.0***
Relevant Coursework: Deep Learning, Advanced Python for Data Science, High-Performance ML, ML for Cyber Security, MLOps

Veermata Jijabai Technological Institute — **08/2016 – 09/2020**
Bachelor of Technology, Electronics and Telecommunication — ***GPA: 8.11/10***

SKILLS

Programming	Python, R, C, C++, SQL, PySpark, Hadoop, MATLAB, Numpy, Pandas, Scikit-Learn, PyTorch, TensorFlow, CUDA
Data Analytics	SVM, k-means, XGBoost, Hypothesis Testing, A/B Testing, LLMs, Time-Series Analysis, Jupyter, RNNs, LSTMs
Software Tools	Azure Databricks, Azure Data Factory, Microsoft Excel, GitHub, PowerBI, Tableau, Grafana, Hive, Docker

WORK EXPERIENCE

Floodnet | *Sensor Data Engineer | NYC, USA* — **04/2024 – 12/2024**
- Created a comprehensive NYC-wide flood sensor health monitoring autonomous system for time series data analysis for **200+ IoT flood sensors** for a $7 million city-wide flood sensing initiative. Reduced downtime by **30%**
- Devised **filters** and **machine learning models** for noise filtering and accurate flood detection **improving accuracy by 6%**
- Collaborated with interdisciplinary teams and contributed to product roadmaps for system enhancements

Fractal Analytics | *Data Science Consultant | Mumbai, India* — **09/2020 – 07/2023**
- Employed advanced analytics on big data with **Python** and **PySpark** on **Azure Databricks** improving performance by **5%**
- Conducted exploratory analysis and designed **ETL pipelines** on **Azure Data Factory** for data processing from ingestion to reporting, utilizing **Cloud Computing** and resulting in a **15% reduction** in overall processing time while enhancing reliability
- Worked on model building, model **validation**, model **deployment**, and **cloud deployment** for the **AI/ML** solutions

PROJECTS

Efficient Vision Transformers: Once-Train Optimization: ***New York University.*** — **09/2024 – 12/2024**
- Developed a once-trained (OT) network framework for **Vision Transformers** (ViTs) to enable efficient deployment across diverse hardware platforms, utilizing **FlashAttention**, **Neural Architecture Search**, **Pruning, Distillation, and Quantization**
- Conducted in-depth **profiling** of **GPU-accelerated applications** to identify performance **bottlenecks**, memory usage patterns, and computational inefficiencies along with **benchmarking** methodologies to measure **inference performance**
- Achieved a model with **95%** fewer parameters and **75%** lower memory used for just a 12% drop in accuracy on CIFAR-100

Continual Learning for Autonomous Vehicles: ***New York University.*** — **01/2024 – 05/2024**
- Developed a **CNN continual learning** system for autonomous vehicle steering control using lifelong learning techniques
- Formulated a novel method of **Temporal Consistency Regulation** with a balanced buffer to enhance prediction accuracy along with smoothness of drive reducing mean square error by **18%** while also reducing memory used by **25%**

Optimized Graph Neural Networks for Credit Card Fraud Detection: ***New York University.*** — **01/2024 – 05/2024**
- Implemented and optimized a **Temporal Graph Attention (TGAT)** mechanism for semi-supervised credit card fraud detection, achieving an **F1-score of 0.88** on the Amazon fraud dataset
- Developed an architecture combining categorical and numerical attributes, using **multi-head attention** for neighbor importance and **attribute-driven gated residuals** to enhance interpretability and performance
- Reduced data preprocessing time by **90%** and model training time by **22%** using **cuda-numba** 1024 threads

Complete Market Solution for Fast Moving Consumer Goods: ***Fractal Analytics.*** — **11/2022 – 07/2023**
- Built **Machine Learning models** for calculating the effects of Innovation of new products into the market, demand transference due to delisting of products, and renovation of products with more than **70% accuracy**
- Predicted annual sales of products using **Random Forest Regression** and **NLP** with **84% accuracy**
- Combined with the pricing tool to create **interactive dashboards**, **simulators**, and **optimizers** to visualize the model outputs and calculate **forecasts** giving product recommendations and enabling fact-based decision-making

ROI calculation based on Bayesian Belief networks: ***Fractal Analytics.*** — **06/2021 – 06/2022**
- Computed feature impact of different variables like Price, Discount, Advertising, and Seasonality on the sales of products, ascertaining which parameters, directly and indirectly, affect sales using **Bayesian Belief Network modeling** in **R**
- Utilized model coefficients to calculate the **Return on Investment** (ROI) for the advertisement media using statistical methods
- Built multiple tools for **investment optimization** resulting in a **7% increase** in ROI

HONORS AND CERTIFICATIONS

- Star Award - Eureka category for developing an innovative solution, *Fractal Analytics*
- Microsoft Certified: Azure Data Engineer Associate
- Winner AI Hackathon, *Symbiosis Institute of Technology, Pune*